CHAPTER 3

ORGANISATION OF DATA

CHAPTER ESSENCE AT A GLANCE:
Transform raw unorganized data into meaningful statistical series: classification principles, discrete versus continuous variables, tally markings, and frequency distributions.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Data Classification Objectives:** Raw data is voluminous and unmanageable; classification condenses data, highlights similarities, facilitates comparison, and prepares data for statistical tabulation.

• **Four Modes of Classification:** Chronological (classified by time), Spatial or Geographical (classified by location), Qualitative (classified by attributes like gender or literacy), and Quantitative (classified by numerical magnitudes like income or age).

• **Discrete vs Continuous Variables:** Discrete variables jump by exact finite units (number of children, bank accounts); Continuous variables assume any fractional value within an interval (height, weight, temperature).

• **Frequency Distribution Mechanics:** Raw numbers are grouped into class intervals (exclusive or inclusive method) using tally bars; midpoint of each class is calculated as upper limit plus lower limit divided by two.

ESSENTIAL ECONOMIC MODELS & GOVERNING FORMULAS

$$\text{Class Midpoint } (m) = \frac{\text{Lower Class Limit } (L_1) + \text{Upper Class Limit } (L_2)}{2}$$

Class Mark and Mid-Value Formula

$$\text{Class Width or Magnitude } (h) = \text{Upper Limit } (L_2) - \text{Lower Limit } (L_1)$$

Class Interval Size Equation

$$\text{Relative Frequency} = \frac{\text{Class Frequency } f_i}{\sum f_i} = \frac{f_i}{N}$$

Proportional Probability Distribution Measure

$$\text{Continuous Variable Invariant} \iff \text{Value can assume any real number in continuous interval } [a, b]$$

Continuum Measurement Property

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** In the Exclusive method (e.g. 10-20, 20-30), the upper limit 20 is excluded from the first class and included in the next class 20-30.

• **Alert:** In the Inclusive method (e.g. 10-19, 20-29), both upper and lower limits are included; it must be converted to exclusive form (9.5-19.5, 19.5-29.5) before computing medians and modes.

• **Alert:** Open-ended classes (e.g. 'Below 10' or '50 and above') lack explicit boundaries, creating difficulties when calculating the arithmetic mean.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. What is meant by Classification of data? State its primary objective. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** The process of arranging raw statistical data into groups or classes according to their common characteristics and affinities. **[1 Mark]**

• **Primary Objective:** To condense vast raw data into manageable forms and highlight points of similarity and difference for easy comparison. **[1 Mark]**

2. Distinguish between a Discrete Variable and a Continuous Variable with one example of each. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Discrete Variable:** A variable that increases by complete discrete jumps and cannot take fractional values (e.g. number of students in a class). **[1 Mark]**

• **Continuous Variable:** A variable that can assume any numerical value within a given specified range, including fractions (e.g. height, weight, body temperature). **[1 Mark]**

3. Explain the four primary bases of Classification of Data with examples. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Chronological Classification:** Data categorized according to time periods like years, months, or quarters (e.g. national income from 2010 to 2024). **[0.75 Mark]**

• **Spatial or Geographical:** Data grouped according to geographical boundaries like states or countries (e.g. wheat production across Indian states). **[0.75 Mark]**

• **Qualitative Classification:** Data classified by non-measurable attributes like gender, literacy, or marital status. **[0.75 Mark]**

• **Quantitative Classification:** Data classified according to numerical magnitudes of a variable like income, height, or marks. **[0.75 Mark]**

4. Distinguish between the Exclusive Method and the Inclusive Method of grouping class intervals. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Exclusive Method:** The upper limit of one class is the lower limit of the next class (e.g. 10-20, 20-30); an observation equal to 20 is included in 20-30, not 10-20. **[1.5 Marks]**

• **Inclusive Method:** Both lower and upper limits are included within the same class (e.g. 10-19, 20-29); an observation equal to 19 is included in 10-19. **[1.5 Marks]**

5. Convert the following Inclusive series into an Exclusive continuous frequency distribution: 10-19 (Frequency: 5), 20-29 (Frequency: 8), 30-39 (Frequency: 12), 40-49 (Frequency: 7). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Correction Factor:** Lower limit of 2nd class (20) minus upper limit of 1st class (19) = 1. Adjustment = 1 divided by 2 = 0.5. **[1 Mark]**

• **Adjustment Rule:** Deduct 0.5 from each lower limit and add 0.5 to each upper limit. **[1 Mark]**

• **Exclusive Series:** 9.5-19.5 (Freq 5), 19.5-29.5 (Freq 8), 29.5-39.5 (Freq 12), 39.5-49.5 (Freq 7). Total N = 32. **[1 Mark]**

6. Explain what is meant by a Cumulative Frequency Distribution. Distinguish between 'Less than' and 'More than' cumulative series. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Concept:** A series obtained by sequentially summing frequencies from the top down or bottom up. **[1 Mark]**

• **Less Than Series:** Frequencies are accumulated starting from the lowest class upward, linked to upper class limits. **[1 Mark]**

• **More Than Series:** Frequencies are accumulated starting from total N downward, linked to lower class limits. **[1 Mark]**

7. Construct a continuous frequency distribution with class intervals of width 10 from the following marks obtained by 20 students: 12, 25, 34, 18, 42, 28, 37, 19, 45, 23, 31, 29, 14, 38, 22, 47, 33, 26, 17, 39. Show tally marks. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Class 10-20:** Marks: 12, 18, 19, 14, 17. Tally: |||| (5 tally bars). Frequency = 5. **[1 Mark]**

• **Class 20-30:** Marks: 25, 28, 23, 29, 22, 26. Tally: |||| | (6 tally bars). Frequency = 6. **[1 Mark]**

• **Class 30-40:** Marks: 34, 37, 31, 38, 33, 39. Tally: |||| | (6 tally bars). Frequency = 6. **[1 Mark]**

• **Class 40-50:** Marks: 42, 45, 47. Tally: ||| (3 tally bars). Frequency = 3. Total Frequency N = 20. **[1 Mark]**